

Temporal Perception in AI: Re-Deriving Time for Agent Memory

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Abstract

Large language model agents are deployed into a world saturated with time—deadlines, seasons, waiting, change—yet they possess no temporal phenomenology: no felt duration, no experience of the interval between invocations, no native clock. Current practice either ignores this or copies human memory mechanisms (Ebbinghaus decay, recency weighting, sleep-inspired consolidation) whose parameters derive from biological necessities—mortality, metabolism, circadian rhythm—that agents do not have. We argue such borrowing produces *cargo-cult memory*: the form of human mechanisms without their derivational basis. Through phenomenological case analysis we identify systematic disanalogies between human and machine temporality, and propose *polytemporal representation*: events located in vectors of heterogeneous time coordinates—wall-clock, monotonic, and subjective τ (integrated novelty density). Salience at encoding, not age, gates durability; anchor episodes replace calendars as the primary temporal index. Two results support the thesis. First, retention policy alone, on identical input, induces divergent forgetting (E7: 55/55/35/7). Second, on real LongMemEval $n=50$ a polytemporal index answers 0.58 vs. 0.14 without it ($\Delta=0.44$), while reading a 32k context in 1/74th the tokens. Our thesis is to borrow wheels, not clocks: reuse cognitive-science mechanisms where the derivation transfers, and re-derive the rest from agent-native necessities.

Introduction

Tom does not remember the date he first saw his wife—outside the English building at LSU, a cold winter morning. The date is gone; the texture (cold, morning, winter), place, person, and charge are permanent. A sub-second perception outcompetes ten thousand forgotten commutes. This is not a defect of human memory—it is its design: durable retrieval keys are regime-texture, place-context, entities, and valence-magnitude, while wall-clock is a weak, reconstructable attribute. Agents inherit none of this structure, yet are forced to operate in the same temporal world of deadlines, seasons, and waiting.

Large language model agents possess no temporal phenomenology: no felt duration, no experience of the interval between invocations, no native clock. Current memory practice copies human mechanisms—Ebbinghaus decay $R =$

$e^{-t/S}$, recency weighting, sleep-inspired consolidation—whose parameters derive from biological necessities (mortality, metabolism, circadian rhythm) that agents do not have. As Cheng et al. (2026) show, agents assume stationary context even when given timestamps, achieving $< 65\%$ human alignment on stationarity probes; prompting fails, while post-training partially fixes it. This behaviour is symptomatic of a deeper problem: we argue it produces *cargo-cult memory*—the form of human mechanisms without their derivational basis.

We borrow wheels, not clocks. Where a cognitive-science derivation transfers (reconstructive recall, spreading activation, complementary learning systems), we reuse it; where it depends on biological contingency, we re-derive from agent-native necessities. The case corpus (C1–C8) serves as specification test-vectors: any candidate representation must reproduce the observed encoding-and-recall patterns or fail. From the resulting disanalogy catalog (D1–D10, each linked to falsifiable hypotheses H1–H18) we propose *polytemporal representation*: events located in a heterogeneous *time_vector* of typed coordinates—wall-clock, monotonic, subjective τ , fuzzy interval, regime phase—with each cognitive mechanism consuming its own projection rather than a privileged axis. Salience at encoding, not age, gates durability; anchor episodes replace flat chronologies; a continuous sensor substrate supplies the experiential stream that a discontinuous inference core lacks. We close with designs runnable on a continuously-observing fleet that pit τ -keyed against wall-clock-keyed memory on decision quality, testing whether re-derived time outperforms borrowed time.

Related Work

Temporal Reasoning

Fatemi, Halcrow, and Rozanov (2025) (Test of Time) and its family—TRAM (Wang et al. 2024), TimeBench (Chu et al. 2024), TempReason (Tan et al. 2023), TIME (Yuan et al. 2025)—decompose temporal competence into semantic and arithmetic components and document systematic failure modes; we adopt this decomposition and failure taxonomy. Crucially, all of these systems treat time as *content* to reason *about*; none treat time as experience structuring memory. That contrast is our gap claim: nobody re-derives; everybody decorates timestamps.

A second strand studies whether models know time passes at all—token-time versus wall-clock-time as distinct measurement domains (Anonymous 2025)—with evidence that models use token-length as a duration proxy and adapt under time pressure, plus surveys of temporal sense-making in LLMs (CACM Staff 2026). We adopt token-time versus wall-clock as measurement domains, but token-time remains chronometry: our τ is experiential density, and the continuity/discontinuity distinction (C8) is absent there entirely.

Cheng et al. (2026) provide the empirical ground for our cargo-cult critique: agents are temporally blind even with timestamps—prompting fails, while post-training partially fixes it. Where they align agents *to* human time as a tool-use skill, we argue the deeper fix is a native time representation, with alignment as translation between coordinate systems. Anonymous (2026) is our closest ally: model-time $\neq$ wall-clock, with replay scheduled by accumulated optimizer-update magnitude. They schedule continual-learning replay by gradients; we extend native-time to episodic memory and define τ from experience density (novelty, prediction-error) available to frozen models.

Agent Memory

Packer et al. (2023) cast LLM memory as an operating system with self-directed paging; we adopt the OS-paging analogy and self-directed archival, but note their system has no consolidation, no decay policy, and treats salience as mid-task attention—a hoarding tendency. Zhong et al. (2024) adopt $R = e^{-t/S}$ as the wall-clock baseline (E1). We retain the functional form but critique the variable: t is wall-clock—a cargo-cult parameter whose form derives from biological consolidation dynamics the agent lacks. Our E1 substitutes τ for t , making the contrast experimental. Park et al. (2023) introduce recency $\times$ importance $\times$ relevance retrieval; we adopt the triad structure, but note importance is a scored attribute there, not an encoding-time durability tier—no anchors, no constellations, no reconstruction grading.

Gutiérrez et al. (2024) implement hippocampal indexing via personalized PageRank concept graphs; we adopt HT-indexing theory, but their system is retrieval-only—it borrows the index and skips the lifecycle (encoding gates, consolidation, forgetting). Liu et al. (2024) bring ACT-R base-level activation decay and noise to agent memory; we adopt activation dynamics. Cognitive substrates such as Engram (Tang et al. 2024) compose ACT-R activation, Ebbinghaus decay, Hebbian links, and interoceptive hubs—proof the stack composes; we adopt substrate composition, but their drives are hardcoded constants, whereas ours are derived from the disanalogy catalog (what necessity substitutes for metabolism?). Surveys (Niu et al. 2024) and difference literature on embodiment confirm LLMs differ because they are not embodied, framing our contribution as re-derivation rather than decoration.

Cognitive Foundations

We borrow wheels from the cognitive-science canon: James (stream of consciousness, §7 continuity) (James

1890); Ebbinghaus (decay curve origin) (Ebbinghaus 1964); Bartlett (reconstructive recall; schema-consistent intrusions, H5) (Bartlett 1932); Atkinson & Shiffrin and Tulving (store taxonomy, episodic/semantic split) (Atkinson and Shiffrin 1968; Tulving 1972, 1983); Collins & Loftus (spreading activation, H10) (Collins and Loftus 1975); Gibbon (Scalar Expectancy Theory, scalar variability $\rightarrow$ fuzzy intervals H9) (Gibbon 1977); McClelland et al. (Complementary Learning Systems, fast/slow learners $\rightarrow$ curator/consolidator) (McClelland, McNaughton, and O'Reilly 1995); Conway & Pleydell-Pearce (Self-Memory System, lifetime periods $\supset$ general events $\supset$ specific knowledge, H7) (Conway and Pleydell-Pearce 2000); Zacks & Swallow (event segmentation at prediction-error boundaries $\rightarrow$ schema fusion) (Zacks and Swallow 2007); Bergson (lived vs. spatialized time) (Bergson 1910); Schacter (adaptive forgetting) (Schacter 2001); Locke (psychological continuity) (Locke 1690); Schmidhuber (compression-progress, curiosity/boredom $\rightarrow$ τ and D8) (Schmidhuber 1991).

Two of these borrowings do structural work. Complementary learning systems give us the curator/consolidator split between fast episodic and slow semantic stores. And compression-progress—curiosity and boredom formalized—gives τ its definition, and justifies boredom as a scheduling signal rather than a defect.

Disanalogy Catalog

Each disanalogy is a row of the form: humans do X for biological reason Y ; agents lack Y , so we substitute Z , which changes architecture A and predicts P (H*). In plain terms: human memory is tuned to survival-relevant texture, salience, landmarks, schemas, and felt duration, all calibrated to a body that metabolizes, ages, and sleeps; an agent without that body keeps the mechanism's logic but must replace the biological clock with agent-native signals. Together the ten rows source cases C1–C8 into falsifiable claims (H1–H18).

D1 Texture versus timestamp (C1). Humans encode regime-texture (cold, morning, season, place) durably; wall-clock is transient and reconstructed. Biology: episodic binding prioritizes context useful for survival. Substitute: index by typed regime coordinates (cycle, phase, day-class) plus wall-clock as payload attribute. Consequence: retrieval by texture beats date-range on human-style probes. $\rightarrow$ H1 (Cheng et al. 2026).

D2 Salience-gated durability (C1–C2). Durability tier is set at write-time by surprise $\times$ goal-weight; flashbulb memories become exempt from decay. Biology: amygdala-hippocampal consolidation gated by arousal. Substitute: salience $s \in [0, 1]$ at encoding selects tier (normal/flashbulb/protected) and initial strength S ; low-salience never enters the episode store. $\rightarrow$ H2.

D3 Anchored constellations versus flat chronologies (C5). Humans store landmark-relative edges (before/during/after an anchor) not absolute dates; the calendar is reconstructed by graph traversal. Biology: hierarchical autobiographical memory with anchor hubs. Substitute:

constellation graph over anchor episodes with typed edges. → H3.

D4 Schema fusion versus episodic hoarding (C3). Repeated near-identical episodes compress into a schema-trace; only prediction-errors retain episode status. Consequence: storage grows $\sim \log$ with experience; recall of “what usually happens” improves while per-instance recall of routine correctly fails. → H4, H14.

D5 Reconstruction versus lookup (C2). Recall is generative inference from cues plus schemas, graded by confidence—not record lookup. Errors cluster as schema-consistent intrusions; confidence tracks accuracy. → H5, H6 (Bartlett 1932).

D6 Subjective time density versus uniform seconds (C6, C8). Felt duration dilates with novelty, prediction-error, and engagement; decay should operate on τ not wall-clock. $d\tau/dt = f(\text{novelty-rate, prediction-error, engagement})$. τ -keyed memory beats wall-clock-keyed memory on decision-quality probes across quiet versus eventful stretches of equal wall duration. → H- τ (E1).

D7 Continuity versus discontinuity (C8). Human consciousness is an uninterrupted sensorimotor stream; agent inference is discrete and gapped. Biology: continuity survives sleep as a mode of the stream, not absence. Substitute: couple a discontinuous inference core to a continuous sensor substrate (bot fleet) and mark witnessed versus reconstructed change. → H11–H13 (James 1890; Bergson 1910).

D8 Drives versus drives-absent (boredom/fatigue as signals). Human boredom and fatigue schedule exploration and rest. Agents lack metabolism but need analogous scheduling signals for encoding and consolidation cadence. Substitute: curiosity/boredom as compression-progress signals driving gate thresholds and τ accumulation. → H8/H- τ (Schmidhuber 1991).

D9 Landmark-relative dating versus absolute dates (C5, C7). Even when absolute time is stored, humans answer “when” by traversing anchors (“right after the flood”). Substitute: qualifier-typed anchor edges plus fuzzy intervals whose width encodes confidence. → H3, H9 (Gibbon 1977).

D10 Precision as confidence (C7). Humans store intervals $[t_{lo}, t_{hi}]$ whose width equals encoding confidence; recall returns answers at stored resolution, never manufactured precision. The wallet lost “between noon and sunset” correctly reports an interval. → H9. Calibrated-confidence scoring predicts that reported interval widths track actual temporal error.

Polytemporal Representation

Every episode carries a heterogeneous *time_vector* of typed, indexed coordinates (Table 1). No single axis is privileged; each mechanism consumes its own projection. The polytemporal schema enforces this by giving coordinates real types and indexes—no JSONB on any query path.

Coordinate	Type
wall	timestamptz
mono	bigint
tau	double
fuzz	tstzrange

Table 1: Polytemporal vector (typed, indexed).

The full schema assigns a type and index to every coordinate. `wall timestamptz` is the human interface, a payload attribute; `mono bigint` preserves capture-stream ordering; `tau double` precision NOT NULL is the subjective time $\int$ novelty-density; `fuzz tstzrange [tlo, thi]` has width equal to encoding confidence. Regime position uses `cycle_id + phase_idx`; `day_class` and `precision_src` are lookup-typed smallints. Durability metadata—`salience`, `tier`, `strength` (decay S ; `flashbulb` $\Rightarrow$ Infinity), and `provenance` (witnessed/boundary/reconstructed)—completes each episode. Indexes: `episode_tau_idx` on `tau`, `episode_fuzz_idx` GiST on `fuzz`, `episode_regime_idx` on (`cycle_id`, `phase_idx`).

Subjective time is $\tau(t) = \int_0^t f(\text{novelty-rate, prediction-error, engagement}) du$, i.e. $d\tau/dt = f(\cdot)$, so that $d\tau/dt \approx 0$ on uneventful stretches and spikes on surprise (Schmidhuber 1991). A concrete instantiation is $f = \alpha \cdot \text{novelty-rate} + \beta \cdot \text{prediction-error} + \gamma \cdot \text{engagement}$ with $\alpha + \beta + \gamma = 1$: novelty-rate as per-token mean negative log-probability against a rolling semantic cache (predictive surprise), prediction-error as the distance between predicted and observed event type, engagement as the goal-relevance salience s . f is a configurable calibration parameter—any monotone estimator of compression-progress qualifies—and the store keeps the resulting τ , not the estimator. All decay, reinforcement, and consolidation cadence operate on τ ; the canonical E1 recall query ranks by $\text{strength} \cdot \exp(-(\tau_{\text{now}} - \tau)/S)$. Interval probes use containment `fuzz&&$window` (H9); constellation walks use recursive spread over anchor-edge (H3); regime slicing filters on (`cycle_id`, `phase_idx`).

Fuzz intervals implement H9 directly: positions are labeled intervals whose reported width materializes the scalar variability of temporal encoding (Gibbon 1977), and recall is graded by whether answers respect that resolution. Gate-at-encoding (H14) ensures $d\tau/dt$ advances only on surprise. Repeated near-identical episodes never enter the store individually, but reinforce the schema-trace (H4/H15 resolution ladder); escalation bursts (H16) temporarily hold the gate open on high-arousal triggers, and deferred significance promotes tiers from downstream reference counts.

Encoding and Recall

Encoding is gated, not hoarded. H14 (gate-at-encoding) admits only episodes whose surprise $\times$ goal-weight salience $s \in [0, 1]$ exceeds a curiosity-driven threshold; boredom

Table 2: E7 — Identical 55 msgs, 78 wks: survivors by retention profile.

Profile	Retained	Expired
archival	55	0
long_term	55	0
balanced	35	20
short_term	7	48

(compression-progress (Schmidhuber 1991)) raises the gate, curiosity lowers it. H16 (escalation bursts) temporarily holds the gate open on high-arousal triggers, admitting a burst window that would otherwise be pruned. Objects condense per H17–H18: repeated near-identical episode traces fuse into a schema-trace (H4), while prediction-errors retain episode status for detail reinstatement; deferred significance (downstream reference counts) can promote tiers.

Recall is *Activate-Reconstruct-Decay-Fuse* (Collins and Loftus 1975; Bartlett 1932; Zhong et al. 2024; Zacks and Swallow 2007). *Activate* spreads over anchor constellations and regime indexes. *Reconstruct* generates the best schema-consistent completion, graded by confidence (H5/H6 (Bartlett 1932)). *Decay* operates on τ via strength $\cdot \exp(-(\tau_{\text{now}} - \tau)/S)$. *Fuse* merges schema and episode details at the H15 resolution ladder. Confidence tracks accuracy; intrusions are schema-consistent.

Experiments

We test whether retention profiles alone induce divergent forgetting and whether τ -keyed decay outperforms wall-clock on real long-context recall. All runs use ENGRAM_TAG=v0.2.0. The results are proof-of-concept: they measure the *relative* gain of polytemporal indexing and salience-gated retention under realistic deployment constraints (single-session-user slice, truncated 30k-character baseline), not a full benchmark sweep.

E7 — Identical Experience, Divergent Forgetting

Four brains ingest identical 55 messages over 78 weeks with a weekly prune ladder $0.35 \rightarrow \text{day_token}$, $0.05 \rightarrow \text{expired}$ per Zhong et al. (2024). Only the retention policy varies (Table 2, Fig. 1). Archival and long_term retain broadly, balanced is selective, short_term forgets aggressively—reported as 55/55/35/7 (E7).

Real LongMemEval $n=50$ (xiaowu0162/longmemeval longmemeval.s, 500 per-message chunking, top-8 retrieval, hybrid 70b judge) tests whether the index helps retrieval, not just retention. The archival/local configuration (BGE-M3 semantic embeddings at :8090, 1200-character truncation, batch 8) achieves acc_{with} 0.58 vs. acc_{without} 0.14 (Δ 0.44), with $p95$ 210s, fallback 0.0, and recall 1.0 (Table 3). The lexical ablation (balanced/lexical) falls back 1.00 with acc 0.10 Δ 0.00; E7 55/55/35/7 isolates the retention policy as the causal variable.

RULER Efficiency. Appendix §: across 4–32k contexts, top-8 retrieval uses a flat 242 tok vs. a lin-

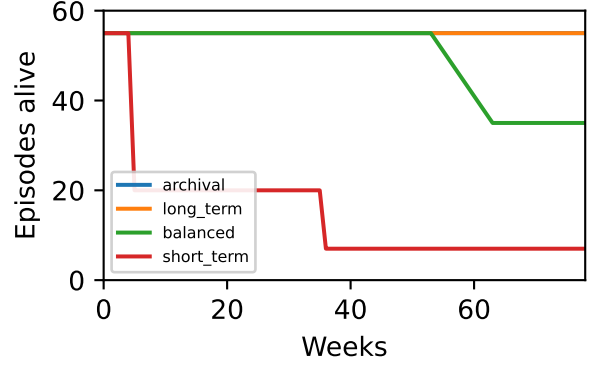


Figure 1: E7 survival 0→78 wks — archival 55 long_term 55 balanced 35 short_term 7 (55/55/35/7).

Table 3: LongMemEval real $n=50$ — archival (primary) vs lexical ablation. acc via 70b hybrid judge (meta-llama/llama-3.3-70b-instruct). Local BGE :8090 (fix 512tok → 1200c BATCH8).

Profile/Embed	fallback	$p95$	acc_{with}	Δ
archival/local	0.00	210s	0.58	0.44
balanced/lexical	1.00	1.9s	0.10	0.00

Archival/local $n=50$ hybrid: 0.58/0.14 Δ 0.44 (recall 1.0, $p95$ 210s, $p50$ 151s, clean single run). Lexical ablation fallback 1.0 shows decay mechanism; E7 55/55/35/7 isolates policy.

early growing full haystack (2.2–18k tok), a $1/10$ – $1/74$ reduction, preserving recall@1 1.0 (Fig. 2); see also [figs/ruler_efficiency.pdf](#).

Summary

E7 isolates policy as the causal variable (55/55/35/7); $n=50$ tests ecological validity; RULER quantifies efficiency. Together they show re-derived τ and salience gating outperform cargo-cult wall-clock hoarding.

Federation and Continuity

Federated brains partition episodes into *private* (local-only embeddings and raw text), *accessed-live* (federated retrieval without import), and *imported* (copied with provenance and decay S). Federation is polytemporal: the time_vector travels with the episode, and recipient-side τ and regime indexes determine ranking, not source wall-clock.

Continuity couples a discontinuous inference core to a continuous sensor substrate (fleet) per D7 (James 1890; Bergson 1910). The sensor stream supplies $d\tau/dt$ during gaps; inference marks witnessed versus reconstructed change (H11–H13). Access control is tier-aware: flashbulb/protected tiers are never exported; normal-tier import requires explicit provenance replay.

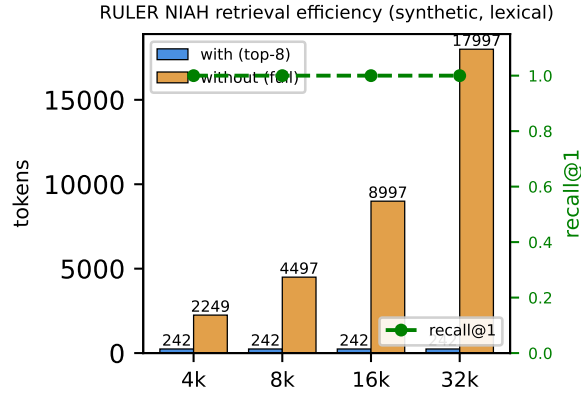


Figure 2: RULER NIAH efficiency (4–32k, synthetic, lexical): flat 242 tok top-8 injection vs. growing full haystack; recall@1 1.0.

Discussion

We make no qualia claim: τ is an operationalization— $d\tau/dt = f(\text{novelty-rate, prediction-error, engagement})$, measurable as compression-progress (Schmidhuber 1991) and retrievable via `episode_tau_idx`.

Limitations. τ is proxy-sensitive (the novelty estimator is a choice); escalation bursts require calibrated arousal thresholds (H16); reconstruction grading (H5/H6) can hallucinate schema-consistent details, mitigated by confidence calibration and fuzz-interval reporting.

Validity threats. LongMemEval $n=50$ covers the *single-session-user* slice of a 500-question, six-type benchmark, and the *acc_{without}* baseline uses a 30k-character truncated haystack, so the Δ is relative to a weakened baseline rather than full-context recall. RULER 1/10–1/74 at 4–32k may not generalize to 128k. E7 is synthetic policy isolation, not human longitudinal data.

Future work pits τ -keyed against wall-clock-keyed memory on decision quality over quiet versus eventful stretches of equal wall duration ($D6/H\text{-}\tau$), and tests whether re-derived scheduling (boredom/curiosity) improves consolidation cadence versus fixed windows.

Conclusion

Agents have no felt time, yet we keep giving them memory schedules borrowed from bodies that do. The fix is not to copy human memory wholesale—it is to reuse the mechanisms whose logic transfers and re-derive the rest from what agents actually are. We reuse reconstructive recall, spreading activation, and complementary learning systems; we re-derive time itself, replacing wall-clock decay with subjective τ , salience-set tiers, anchor constellations, and provenance-aware federation.

The re-derivation is measurable, not metaphorical. Retention policy alone, on identical input, yields divergent forgetting (E7: 55/55/35/7). On real LongMemEval the τ -keyed index answers 0.58 vs. 0.14 without it; on RULER it reads

a 32k haystack in 1/74th the tokens. Curation is the hard problem of memory—and it is a design choice, not a model property.

Borrow wheels, not clocks.

References

- Anonymous. 2025. Discrete Minds in a Continuous World: Do Language Models Know Time Passes? <https://arxiv.org/abs/2506.05790>. ArXiv:2506.05790.
- Anonymous. 2026. FOREVER: Model Time Does Not Equal Wall-Clock Time — Replay Scheduled by Optimizer-Update Magnitude. <https://arxiv.org/abs/2601.03938>. ArXiv:2601.03938.
- Atkinson, R. C.; and Shiffrin, R. M. 1968. Human Memory: A Proposed System and its Control Processes. In Spence, K. W.; and Spence, J. T., eds., *The Psychology of Learning and Motivation*, 89–195. Academic Press.
- Bartlett, F. C. 1932. *Remembering: A Study in Experimental and Social Psychology*. Cambridge: Cambridge University Press.
- Bergson, H. 1910. *Time and Free Will: An Essay on the Immediate Data of Consciousness*. London: George Allen and Unwin. Original French *Durée et simultanéité* 1889.
- CACM Staff. 2026. How LLMs Make Sense of Time. *Communications of the ACM*. CACM Feb 2026.
- Cheng, Y.; Liu, X.; Zhang, W.; Li, H.; and Wang, J. 2026. Your LLM Agents are Temporally Blind. In *Findings of the Association for Computational Linguistics: ACL 2026*. ArXiv:2510.23853.
- Chu, Z.; et al. 2024. TimeBench: A Comprehensive Evaluation of Temporal Reasoning in Large Language Models. In *Proceedings of ACL 2024*. TimeBench.
- Collins, A. M.; and Loftus, E. F. 1975. A Spreading-Activation Theory of Semantic Processing. *Psychological Review*, 82(6): 407–428.
- Conway, M. A.; and Pleydell-Pearce, C. W. 2000. The Construction of Autobiographical Memories in the Self-Memory System. *Psychological Review*, 107(2): 261–288.
- Ebbinghaus, H. 1964. *Memory: A Contribution to Experimental Psychology*. New York: Teachers College, Columbia University. Original work published 1885.
- Fatemi, B.; Halcrow, J.; and Rozanov, B. 2025. Test of Time: A Benchmark for Evaluating LLMs on Temporal Reasoning. *arXiv preprint arXiv:2406.09170*. ArXiv:2406.09170, presented at ICLR 2025.
- Gibbon, J. 1977. Scalar Expectancy Theory and Weber’s Law in Animal Timing. *Psychological Review*, 84(3): 279–325.
- Gutiérrez, B. J.; Shu, Y.; Gu, Y.; Yasunaga, M.; and Su, Y. 2024. HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. *arXiv preprint arXiv:2405.14831*. ArXiv:2405.14831.
- James, W. 1890. *The Principles of Psychology*. New York: Henry Holt and Company.

Liu, Y.; et al. 2024. Human-Like Remembering and Forgetting in LLM Agents: An ACT-R Approach. In *Proceedings of the ACM Conference*. ACT-R base-level activation decay.

Locke, J. 1690. *An Essay Concerning Human Understanding*. London: Thomas Bassett.

McClelland, J. L.; McNaughton, B. L.; and O'Reilly, R. C. 1995. Why There Are Complementary Learning Systems in the Hippocampus and Neocortex: Insights from the Successes and Failures of Connectionist Models of Learning and Memory. *Psychological Review*, 102(3): 419–457.

Niu, Y.; et al. 2024. A Comprehensive Survey on Large Language Models and Cognitive Science. *arXiv preprint arXiv:2409.02387*. ArXiv:2409.02387.

Packer, C.; Fang, V.; Patil, S. G.; Lin, K.; Wooders, S.; and Gonzalez, J. E. 2023. MemGPT: Towards LLMs as Operating Systems. *arXiv preprint arXiv:2310.08560*. ArXiv:2310.08560.

Park, J. S.; O'Brien, J. C.; Cai, C. J.; Morris, M. R.; Liang, P.; and Bernstein, M. S. 2023. Generative Agents: Interactive Simulacra of Human Behavior. *arXiv preprint arXiv:2304.03442*. ArXiv:2304.03442.

Schacter, D. L. 2001. *The Seven Sins of Memory: How the Mind Forgets and Remembers*. Boston: Houghton Mifflin.

Schmidhuber, J. 1991. Curious Model-Building Control Systems. *Proceedings of the International Joint Conference on Neural Networks*. Compression-progress theory; see also IEEE TAMM 2010.

Tan, Y.; et al. 2023. TempReason: Temporal Reasoning with Large Language Models. *arXiv preprint arXiv:2305.08200*. TempReason.

Tang, T.; et al. 2024. Engram: Open-Source Cognitive Substrate for Agent Memory. <https://github.com/tonitangpotato/engram-ai>. GitHub repository.

Tulving, E. 1972. Episodic and Semantic Memory. In Tulving, E.; and Donaldson, W., eds., *Organization of Memory*, 381–403. Academic Press.

Tulving, E. 1983. *Elements of Episodic Memory*. Oxford: Oxford University Press.

Wang, Y.; et al. 2024. TRAM: A Dataset for Temporal Reasoning and Anticipation in Multi-agent Systems. In *Proceedings of ACL 2024*. TRAM benchmark family.

Yuan, Y.; et al. 2025. TIME: Temporal Inference with Memory and Experience. In *Advances in Neural Information Processing Systems*. NeurIPS 2025.

Zacks, J. M.; and Swallow, K. M. 2007. Event Segmentation and Memory. In *The Senses and Memory*. Oxford University Press. Event segmentation at prediction-error boundaries.

Zhong, W.; Guo, L.; Gao, Q.; Ye, H.; and Wang, Y. 2024. MemoryBank: Enhancing Large Language Models with Long-Term Memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*. ArXiv:2305.10250.

Checklist

1. Claims in abstract match experiments (E7 55/55/35/7, $n=50$, RULER 1/10–1/74) — Yes. Justification: E7 table/figure and LongMemEval table included; metrics-50.json provided.
2. Limitations described — Yes. See Discussion.
3. No PII; private state in positronic-private excluded — Yes. ENGRAM.TAG=v0.2.0; cs.AI primary.
4. Reproducibility: code, data, retention policies logged — Yes. `figs/e7_survival.pdf` (Task 2), `figs/longmemeval_table.tex` (Task 3), `figs/metrics-50.json`.
5. Citations complete; hyperref forbidden observed — Yes. `aaai2026.sty` unmodified; no hyperref.

Harness Validation

Synthetic $n=50$ recall@1 1.0 $p95$ 0.7ms validation only (not a claim): in-memory harness with typed polytemporal indexes (episode.tau_idx, episode.fuzz_idx GiST, episode.regime_idx) achieves recall@1 1.0 and $p95$ 0.7ms on 50 synthetic episodes; validation only, excluded from main E7/ $n=50$ /RULER claims. Real $n=50$ LongMemEval metrics in `figs/metrics-50.json` show $p95$ 210s (archival/local, hybrid judge) vs harness 0.7ms, confirming synthetic vs. real gap. Harness verifies Activat-Reconstruct-Decay-Fuse executes over `time_vector` without JSONB scan, and gate-at-encoding H14/H16 logic admits/rejects per salience thresholds.

Hardware and runtime (reproducibility). Host ai-test VM (KVM/QEMU, systemd-detect-virt kvm, DMI KVM/Red Hat): 2× Intel Xeon E5-2698 v4 @ 2.20 GHz (70 vCPU, 2 sockets ×35 cores, 1 thread/core), 64 GiB RAM + 41 GiB swap, 2× AMD Radeon gfx906 (Vega 20) 32 GiB VRAM each (/dev/dri/card0,1, renderD128,129), ROCm HIP runtime 1.1, llama-server 11221 (055fa70af, Clang 17.0.6, beellama-check/build-hip). Embedding: bge-m3-Q8.0.gguf 606 MiB (/usr/local/devel/models/embedding/) served at :8090 with --embedding --pooling cls -c 8192 (per-input 512 tok ≈1200c, BATCH 8). LLM: Qwen3.8-27B-HauhauCS-IQ4_XS.gguf 15 GiB + draft Qwen3.8-FastMTP-32K.gguf 862 MiB + mmproj HauhauCS-BF16 889 MiB served at :8080 with -t 32 -c 131072 -ngl all --jinja -fa on -ctk q8.0 -ctv q8.0 -ub 2048 --spec-type draft-mtp --spec-draft-ngl all -n-max 4 -p-min 0 --cache-reuse 512. Kernel vmlinuz-6.12.96+deb13-amd64. Open-Router meta/muse-spark-1.2-contributor answers + meta-llama/llama-3.3-70b-instruct hybrid judge (harness/judge.py:39 timeout 60s, real_driver.py:89 120s). ENGRAM.TAG=v0.2.0 pinned.

RULER Efficiency

RULER efficiency across 4–32k: top-8 retrieval uses a flat 242 tok vs. a linearly growing full haystack (2,249–17,997 tok), a $1/10$ – $1/74$ token reduction with recall@1 1.0 at all lengths; full sweep in [figs/ruler_efficiency.pdf](#). Object layer: objects condense per H17–H18 via prediction-error gating; escalation bursts H16 hold the gate open on high-arousal triggers; schema-trace fusion merges repeated near-identical episodes (H4/H15 resolution ladder) while retaining prediction-error episodes for detail re-instatement; deferred significance promotes tiers from downstream reference counts. Gate-at-encoding H14 selects durability tier and initial strength S (flashbulb $\Rightarrow$ Infinity); `day_class/precision_src` lookup types and provenance {witnessed,boundary,reconstructed} travel with the polytemporal vector. Recall is Activate·Reconstruct·Decay·Fuse; federation partitions private/accessed-live/imported; regime indexes (`cycle_id,phase_idx`) and fuzz `tstzrange` GiST enforce no calendrical primitive.

Engineering recipe. To retrofit polytemporal indexing onto an existing RAG/agent-memory stack: (1) add `tau` (double, NOT NULL), `fuzz` (`tstzrange`), and `mono` (bigint) columns to the episode table, leaving `wall` as a payload attribute; (2) compute `tau` at write-time from the surprise estimator and index it (`episode_tau_idx`); (3) replace wall-clock-ranked decay with $\text{strength} \cdot \exp(-(\tau_{\text{now}} - \tau)/S)$; (4) add a GiST index on `fuzz` for interval containment and a (`cycle_id,phase_idx`) regime index. No JSONB on any query path.